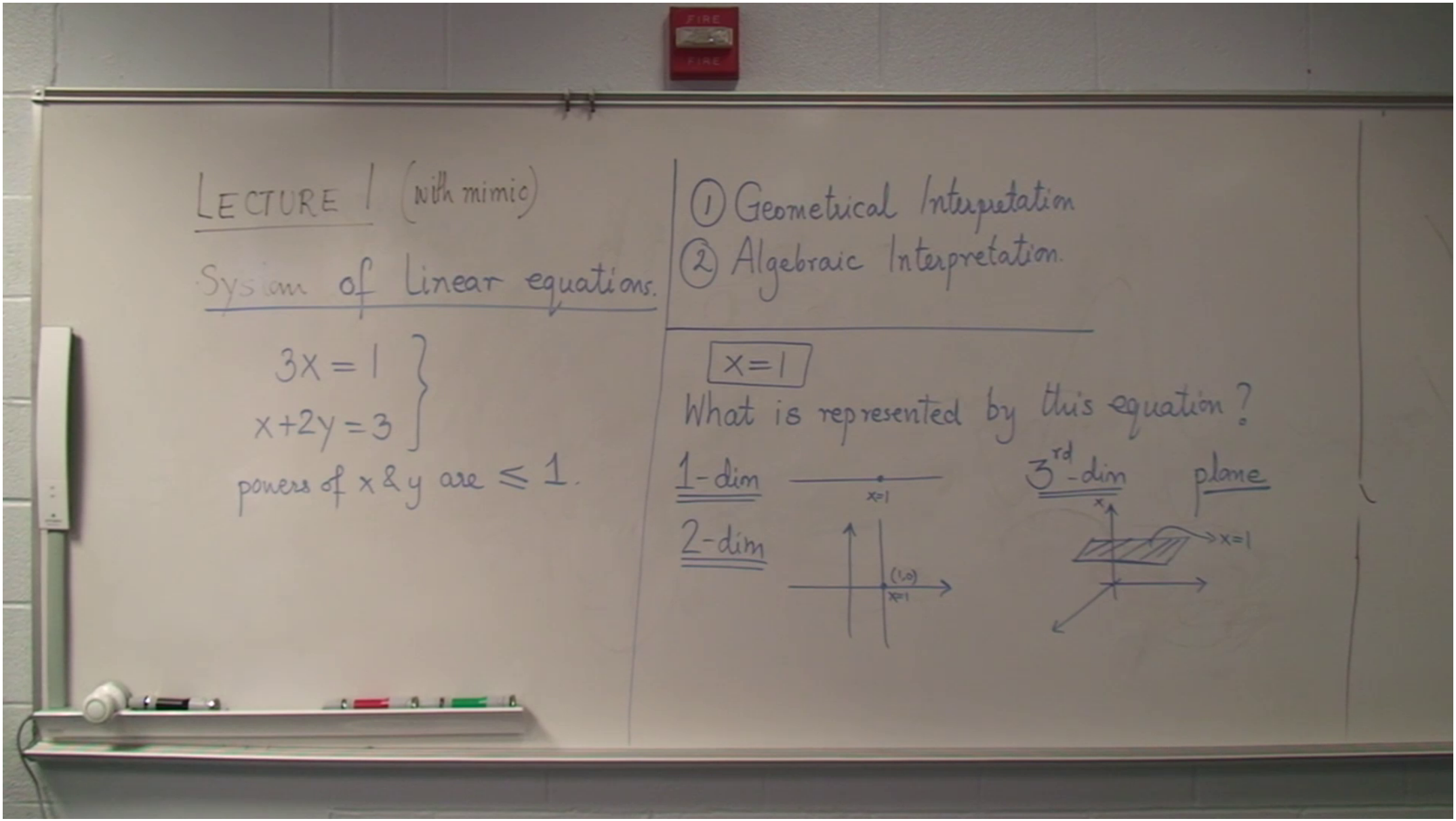
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LECTURE 1 (with mimic)

System of linear equations.

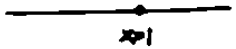
$$\left.\begin{aligned} 3x &= 1 \\ x+2y &= 3 \end{aligned}\right\}$$

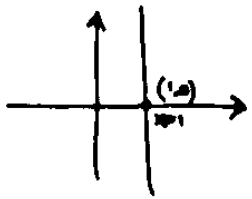
powers of x & y are ≤ 1.

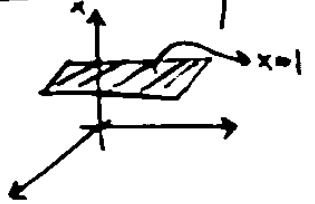
① Geometrical Interpretation
② Algebraic Interpretation.

$x=1$

What is represented by this equation?

1-dim 

2-dim 

3rd-dim plane 

LECTURE 1 (with mimic)

System of linear equations.

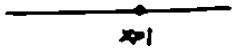
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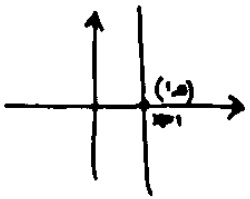
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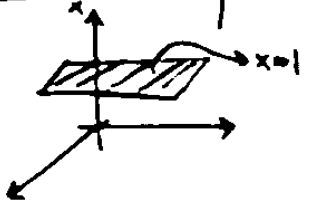
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System of linear equations.

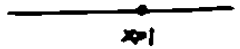
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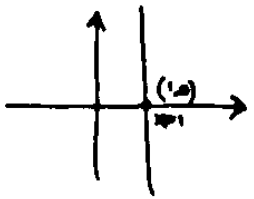
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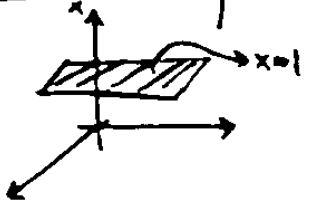
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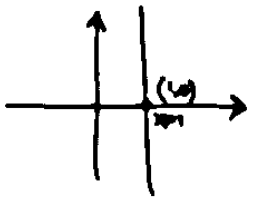
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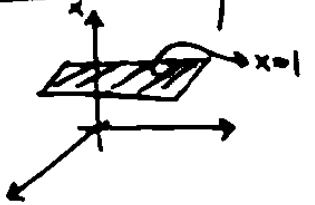
1-dim



2-dim



3rd-dim plane



LECTURE 1 (with mimic)

System of linear equations.

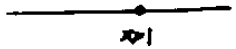
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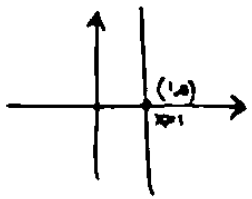
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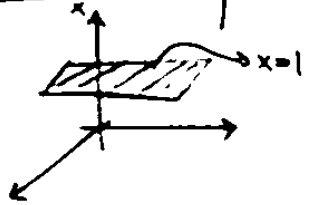
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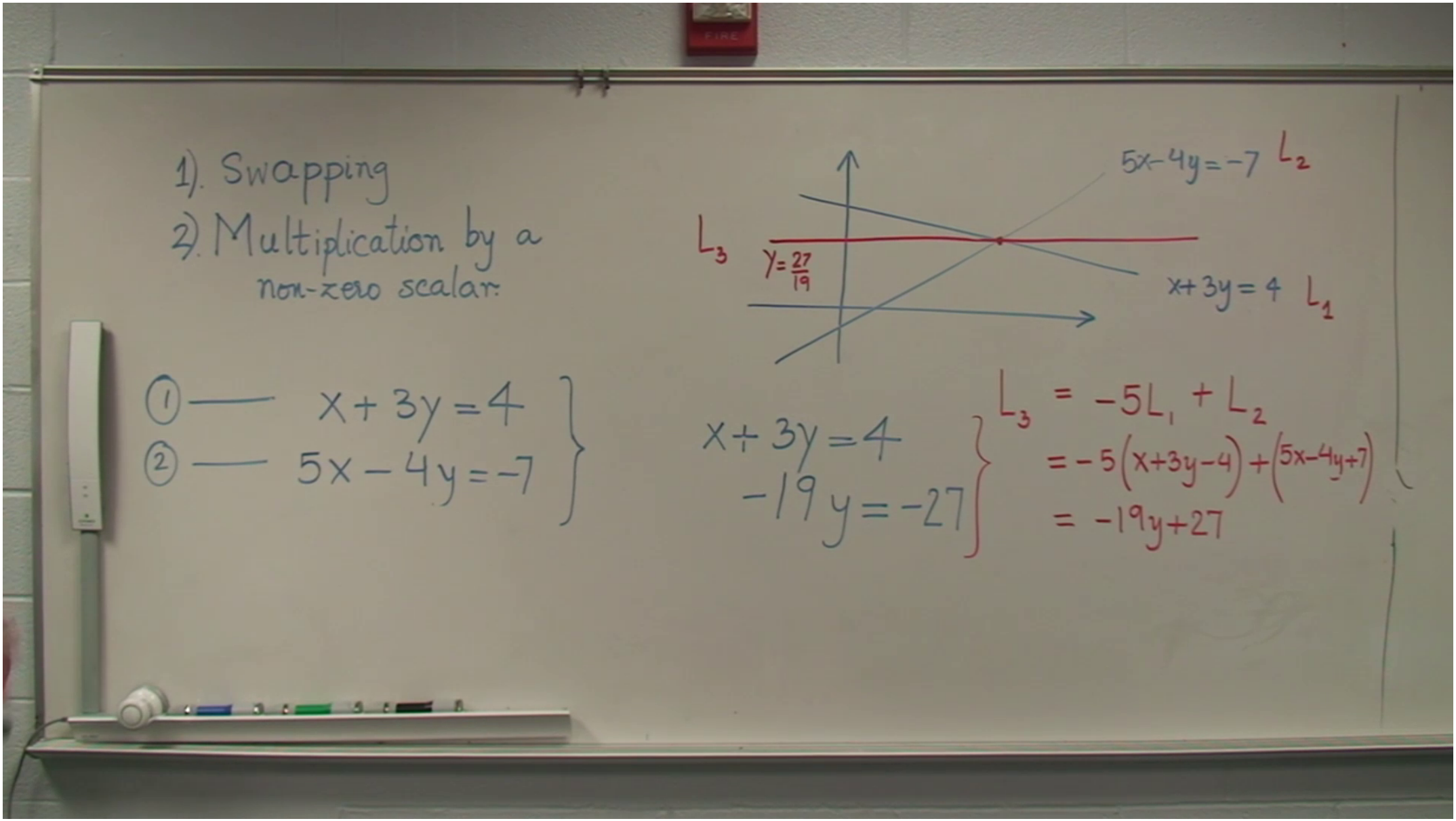
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3rd-dim plane 

REAL lecture_02_44601 | Original RGB | native 1920x1080, no smoothing

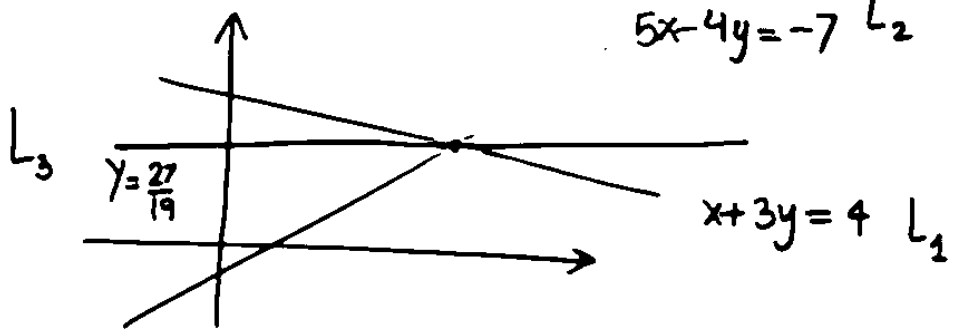


1). Swapping

2). Multiplication by a non-zero scalar.

① —— $x+3y=4$

② —— $5x-4y=-7$



$x+3y=4$

$-19y=-27$

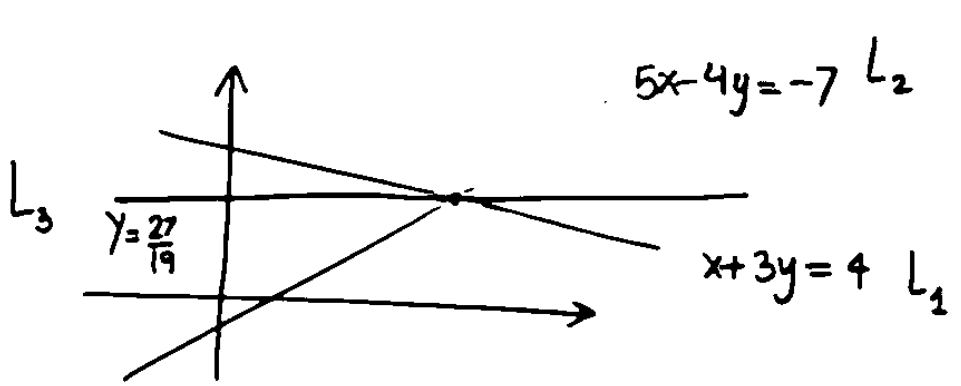
$L_3 = -5L_1 + L_2$

$= -5(x+3y-4) + (5x-4y+7)$

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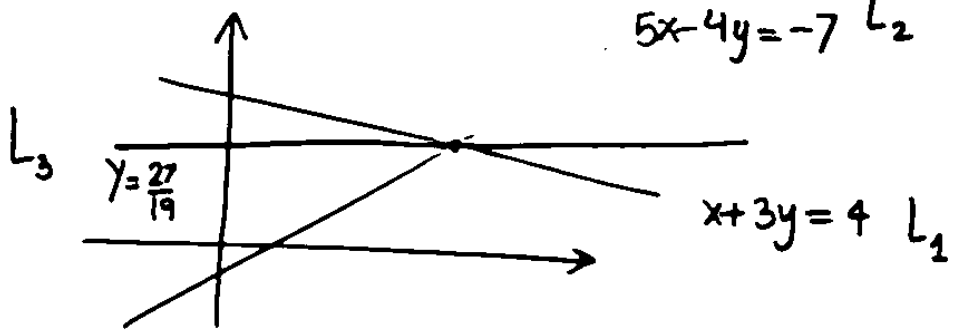
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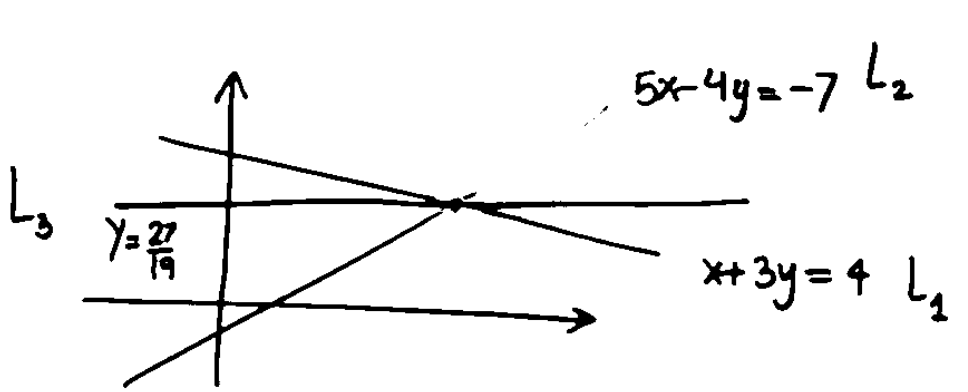
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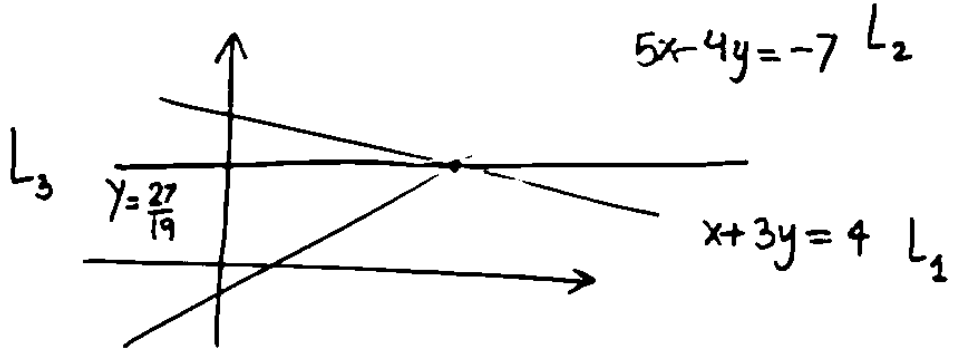
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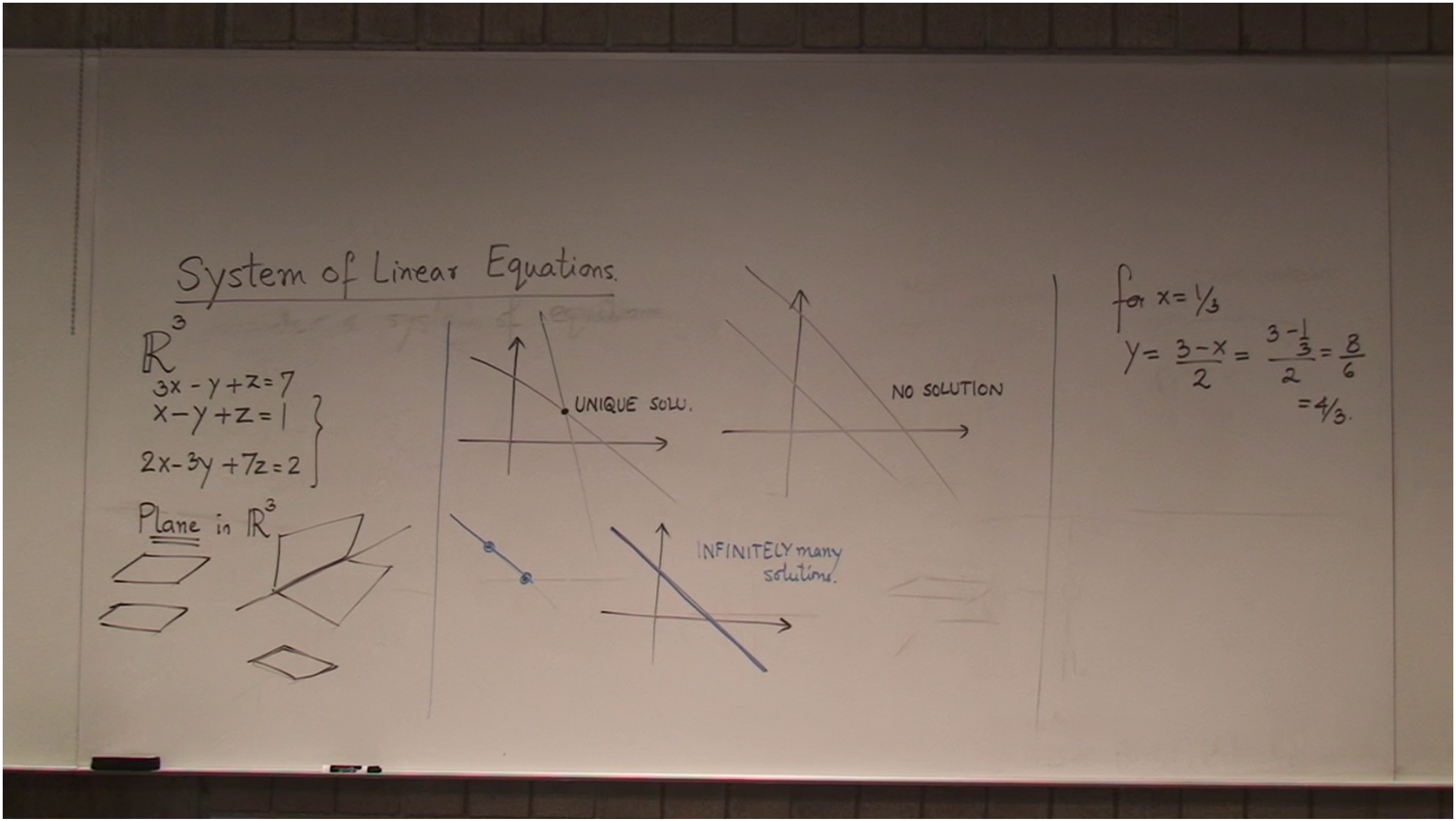
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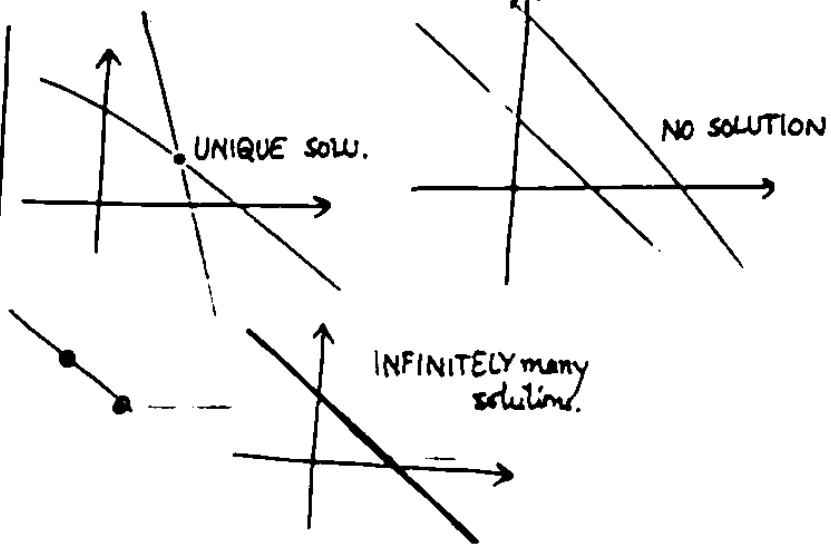
System of Linear Equations.

$\mathbb{R}^3$

$3x - y + z = 7$
$x - y + z = 1$
$2x - 3y + 7z = 2$

Plane in $\mathbb{R}^3$





for $x = 1/3$

$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$

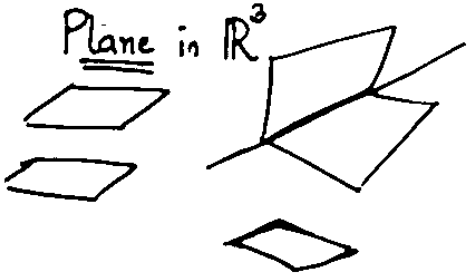
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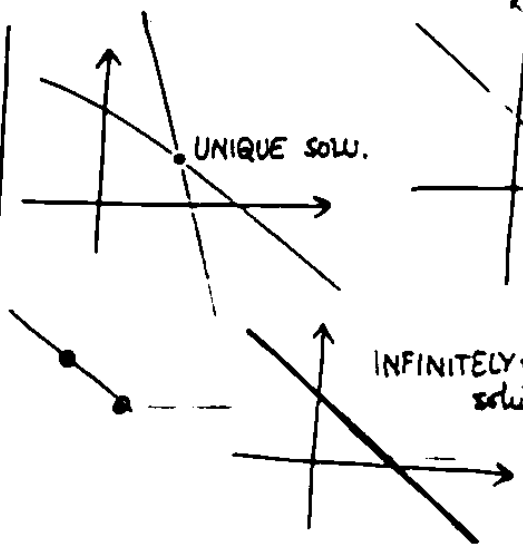
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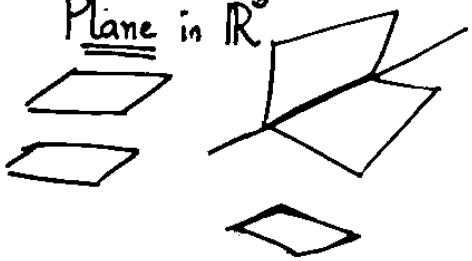
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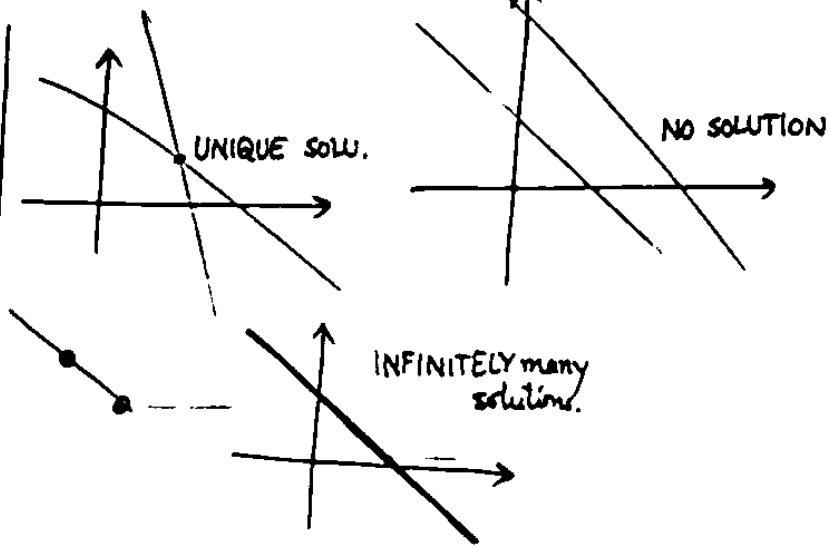
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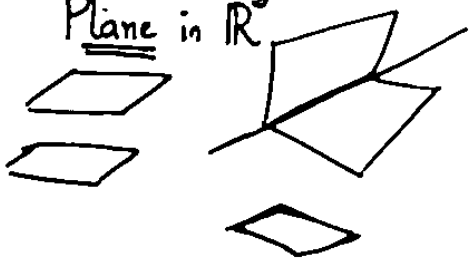
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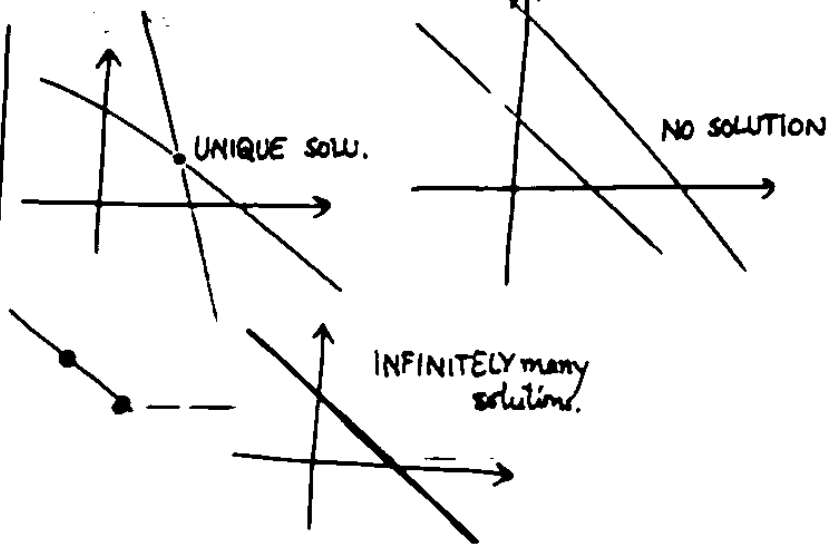
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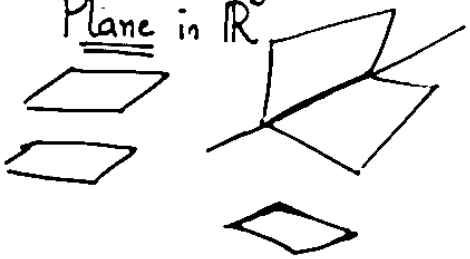
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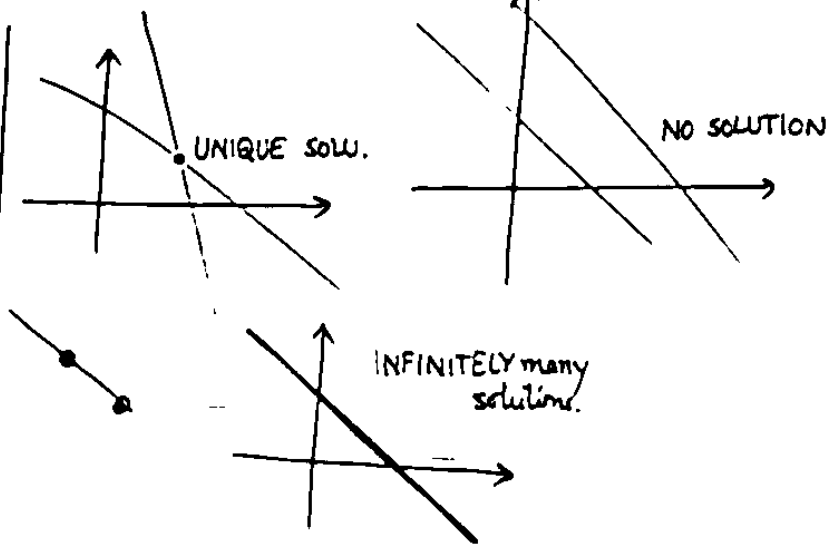
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